Anton Kutyrev, PhD

Lecturer and programme lead in Exploration Geology, Cardiff University Kazakhstan, Astana, Kazakhstan
KutyrevA1@cardiff.ac.uk | ORCID 0000-0001-5933-6455 | antonkutyrev.com

Research focus

Petrology and mineralogy of mafic-ultramafic and island-arc systems; melt-rock and fluid-rock interaction; origin of chromitites; platinum-group elements (PGE) mineralogy from hand sample to nanoscale; new minerals; PGE/PGM cycling and mobility in melts and fluids in ultramafic rocks and layered intrusions; oxygen isotopes as tracers of crustal recycling and early Earth surface environments; critical minerals in the context of the energy transition.

Education

2020 **PhD in Geology and Mineral Sciences.**

Defended in 2019 and awarded in 2020 through the Dissertation Council at the V.S. Sobolev Institute of Geology and Mineralogy (SB RAS); research carried out at the Institute of Volcanology and Seismology (RAS), Petropavlovsk-Kamchatsky, Russia.

2017 **Specialist degree with Distinction in Applied Geology,** Saint Petersburg Mining University, Russia.

Five-year full-time programme, broadly equivalent to a Master of Science.

Academic appointments

2026–present **Lecturer in Exploration Geology,** Cardiff University Kazakhstan, Astana, Kazakhstan. One-year fixed-term appointment.

Teaching, programme leadership, curriculum development and research in economic and exploration geology, including layered intrusions, chromitites, platinum-group elements, critical minerals and isotope geochemistry. Lead the Exploration Geology programme on the Kazakhstan side and coordinate its local delivery; organising the programme's first fieldwork and developing a new field course from scratch. Continuing research in earlier study areas, including the Bushveld Complex, South Africa, and beginning work on the geology of Kazakhstan.

2025–2026 **Postdoctoral Research Fellow,** School of Geosciences, University of the Witwatersrand, Johannesburg, South Africa.

Critical minerals in layered mafic–ultramafic intrusions and Bushveld chromitites within the DSTI CIMERA industry–public consortium, with a focus on Ni-Cu-PGE systems, chromitites and the petrogenesis of the Critical Zone.

2023–2025 **Research Associate,** School of Earth and Environmental Sciences, Cardiff University, Wales, UK.

Critical-mineral and PGE geochemistry in layered intrusions and related volcanic rocks within the Horizon Europe SEMACRET consortium on energy-transition mineral systems, in an interdisciplinary team spanning geology, geophysics and social sciences.

2022–2023 **Fulbright Visiting Scholar,** School of Earth Sciences, University of Oregon, USA.

Early Archean climate from triple oxygen and hydrogen isotopes in 3.8 Ga metalavas of the Saglek-Hebron terrane, including development of isotope fractionation models.

2020–2022 **Research Associate,** Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia.

PGE mineralization and hydrothermal processes in subduction-related magmatic systems of Kamchatka.

2017–2020 **PhD candidate,** Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia.

Geology and platinum mineralization of concentrically zoned dunite–clinopyroxenite–gabbro complexes of the Tamanvayam and Epilchik groups.

2012–2017 **Engineer (0.5 FTE),** Karpinsky Russian Geological Institute, Saint Petersburg, Russia.

Regional geology and petrology, including petrography and geochemical data for igneous and metamorphic rocks; work on rock collections of the Academician F.N. Chernyshev Central Geological Research Museum and preparation of a field rock identification guide.

Industry appointments

2018 **Geologist (summer fieldwork),** Kirganik Ltd. (Amur Minerals subsidiary), Kamchatka, Russia.

Exploration for porphyry Cu and epithermal Au systems: field mapping, alteration and mineralization logging, geochemical sampling.

2016–2017 **Geologist (summer field seasons),** Kamp Ltd., Kamchatka, Russia.

Junior exploration for PGE mineralization: prospecting, field mapping, heavy-mineral sampling, rock and soil geochemical sampling.

2015 **Field technician (summer fieldwork),** Kamchatgeologia, Kamchatka, Russia.

Regional geological mapping at 1:200,000 scale: traverses, outcrop description and systematic sampling for petrography and geochemistry.

Teaching experience

2026–present **Cardiff University Kazakhstan.** Modules taught: Exploring the Planet Earth (Geology 101), Earth Materials, World of Dynamic Environments, and GIS and Fieldwork. Designing a new field course from scratch, including its academic structure, field activities and teaching materials.

2025–2026 **University of the Witwatersrand.** Five-lecture module on magmatic mineral deposits for honours students; demonstrator on the honours field trip (Northern Bushveld).

2023–2025 **Cardiff University.** Practical sessions for the Mineral Deposits course; guest lectures on mineral deposits; demonstrator on field courses in geological mapping, exploration geology and environmental geoscience (Cyprus, Cornwall, South Wales).

2022–2023 **University of Oregon.** Demonstrator on petrology field courses; guest lectures in petrology and geohazards.

Formal teaching training

2024 Teaching to Improve the Quality of Student Learning, Cardiff University.

2024 Lecturing Skills, Cardiff University.

2024 Being a Good LGBTQA+ Ally, Cardiff University.

Student supervision and mentoring

2026–present **Co-supervisor** (with Prof. Roger Gibson) of the project “Calc-silicate xenoliths in the Lower Zone of the Bushveld Complex and their implications for platinum-group element mineralization” (student: Mensacia Phaswane).

2021–present Research mentoring: Nikolay Baykov, research on the Tumrok magmatic system.

2025–2026 Research mentoring: Vishnu, postgraduate research at the University of the Witwatersrand.

2021–2022 Research mentoring: Pavel Zhegunov, mineralogy of the Evevpenta epithermal silver-gold occurrence, Kamchatka.

Additional professional training

2026	PetThermoTools virtual workshop, University of California, Berkeley.
2025	Reading the Rocks, Polokwane, South Africa.
2025	Resource Geology Workshop, GeoCongress 2025, Bloemfontein, South Africa.
2024	MDSG workshop “Magmatic Ni-Cu Mineral Systems in Europe”, Edinburgh, UK.
2021	TESCAN electron microscopy course, Saint Petersburg, Russia.
2018	Gold Short Course, Prague, Czech Republic.

Analytical and modelling skills

Hands-on: SEM-EDS; Raman spectroscopy; field-portable XRF; clean-lab chemistry and mass spectrometry for Rb-Sr, Sm-Nd; triple-oxygen isotopes (laser fluorination line).

Data reduction and interpretation: EBSD; X-ray diffraction; synchrotron micro-XRF/XAS; FIB-TEM; EPMA; atom-probe tomography; LA-ICP-MS.

Modelling and data processing: Python and spreadsheet-based workflows; MELTS- and MAGEMin-based thermodynamic modelling; Monte Carlo modelling of triple-oxygen isotope exchange; mass-balance and mixing models.

Scientific service

2026–present	**Academic advisor** of SGA and SEG Kazakhstan student chapters.
2024–2025	**Coordinator,** Minerals and Energy Research Centre, Cardiff University.
2023	**Organizing committee member,** 14th International Platinum Symposium, Cardiff, UK.
Reviewer	for *Geology*, *Ore Geology Reviews*, *Mineralium Deposita*, *Nature Communications*, *Chemical Geology*, *Journal of Petrology*, *Australian Journal of Earth Sciences*, *Contributions to Mineralogy and Petrology*, *Journal of the Geological Society*, *Lithos* and *Marine Geology*.

Competitively awarded funding

Amounts converted from foreign currencies to EUR; approximate.

2024	**AuScope Access Program** (Curtin University), in-kind analytical access: “Silicate–oxide–platinum group minerals intergrowths: implications for unknown aspects of low-T PGE behaviour”. Approx. EUR 10,700.
2022–2023	**Fulbright Scholarship** (Fulbright Visiting Scholar). Approx. EUR 26,600.
2020–2022	**RFBR Project No. 20-35-70008** (project executor): “Formation and alteration of minerals in active volcanic hydrothermal systems (Kamchatka)”. Approx. EUR 114,000.
2020–2021	**RFBR Project No. 20-05-00290** (principal investigator): “Distribution of Ni-Fe-Co intermetallic compounds and platinum-group minerals in mafic and ultramafic rocks of various formational types”. Approx. EUR 21,500.
2018–2020	**RSCI Project No. 16-17-10145** (project executor): “Melt–fluid evolution, magmatic immiscibility, and metal budgeting in subduction-related magmas”. Approx. EUR 171,000.
2018–2019	**RFBR Project No. 18-35-00126** (principal investigator): “Typomorphism and formation conditions of PGE minerals in concentrically zoned ultramafic massifs (Epilchik area, Koryak-Kamchatka belt)”. Approx. EUR 9,700.
Total	Approx. EUR 353,600 across the projects, scholarship and analytical access listed above, including projects in which I was a team member. PI-led project awards: approx. EUR 31,200.

Field work

2025–2026	Mapping and sampling of the Critical Zone across the western and eastern Bushveld Complex, South Africa (one month of fieldwork plus regular mine visits).

2023–2025	Fieldwork at the Rhiw layered intrusion, Wales, UK (8 days in total).
2021	Principal investigator and organizer of a helicopter-supported expedition (12 participants) to the Tumrok Range arc picrites, Kamchatka (3 weeks).
2020	Sampling of alluvial awaruite in the Kamchatsky Mys ophiolite, Kamchatka (2 weeks).
2020	Documenting cooling lava flows and caves of the 2012–13 Tolbachik Fissure Eruption (2 weeks); observations of Klyuchevskoy volcano.
2019	Underground and field sampling of mineralized intrusions and related flood basalts, Norilsk (2 weeks).
2018	Documenting lava temperatures in the caves of the 2012–13 Tolbachik Fissure Eruption (2 weeks); visual observations of Shiveluch volcano.
2018	Fieldwork on the Kamchatsky Mys ophiolite (2 weeks).
2018	Prospecting at the Kirganik Cu porphyry and epithermal Au occurrence (1 month).
2016–2017	Prospecting for lode and placer PGE mineralization, Epilchik and Matysken areas, Koryak Highlands (5 months in total).
2015	Geological mapping at 1:200,000 scale, Koryak Highlands (3 months).

Professional societies

2026–present	Associate Member (APONEN), Professional Association of Independent Subsoil Experts (PONEN), Kazakhstan's professional association of geologists and mining specialists.
2023–present	Mineralogical Society of Great Britain and Ireland.
2021–present	Society of Economic Geologists (SEG).
2017–present	Society of Geology Applied to Mineral Deposits (SGA).

Awards

2023	Minerals Young Investigator Award.
2021	Marie Skłodowska-Curie Actions Seal of Excellence.

Public outreach

School visits in the UK and Russia for ages 8 to 16: interactive lectures on volcanoes, minerals and rocks, the societal impact of geology, and career perspectives of a geoscientist.

Publications

Peer-reviewed articles

1. **Kutyrev, A.**, Filimonova, O., Saxey, D.W., Vymazalová, A., Evans, K., Kamenetsky, V.S., Wermeille, D., Fougerouse, D., Reddy, S.M., 2026. Formation of natural IrO_2 through Os-Ir-Ru alloy oxidation. *Chemical Geology*, 123167. https://doi.org/10.1016/j.chemgeo.2025.123167
2. Eslami, A., Chardelin, M., O'Driscoll, B., Malvoisin, B., Evans, K., Grieco, G., Bussolesi, M., & **Kutyrev, A.** 2026. Chromitite-mafic dike interfaces as archives of syn-kinematic fluid-rock interaction, chromium mobility and Ca-metasomatism, Nain Ophiolite, Central Iran. *Journal of Petrology*, egag076. https://doi.org/10.1093/petrology/egag076
3. Rhoads, E.A., **Kutyrev, A.**, Bindeman, I.N., Lynn, K.J., Trusdell, F.A., Downs, D.T., Edwards, H.R., Cook, G.W., Day, J.M.D., 2025. Rhenium-osmium and oxygen isotope homogeneity during the 2022 Mauna Loa eruption and implications for basaltic magma storage. *Bulletin of Volcanology* 87. https://doi.org/10.1007/s00445-025-01825-0

4. Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Barnes, S.J., Stepanov, A.S., Konstantinov, S.E., Baykov, N.I., Kryazhev, S.G., Belousov, I.A., Zhukova, I.A., Scherbakov, V.D., Kalugin, V.M., Ivanov, A.V., Bryanskiy, N.V., Park, J.W., 2025. Erratic behavior of platinum-group elements along an olivine-Cr-spinel cotectic in sulfide-undersaturated basaltic magma: roles of Cr-spinel and alloys. *Scientific Reports* 15, 14366. https://doi.org/10.1038/s41598-025-97281-z
5. Zhitova, E.S., Nuzhdaev, A.A., Davydova, V.O., Sheveleva, R.M., Zhegunov, P.S., Kuznetsov, R.A., **Kutyrev, A.V.**, Khokhlova, M.A., Vlasenko, N.S., 2025. Calcareous skarn-like mineral paragenesis from unaltered basalt of the Alaid volcano (Kuril-Kamchatka island arc). *Minerals* 15. https://doi.org/10.3390/min15030237
6. **Kutyrev, A.**, Bindeman, I.N., O'Neil, J., Rizo, H., 2024. Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada: Implications for moderately low-$\delta^{18}O$ Eoarchean ocean. *Chemical Geology* 670. https://doi.org/10.1016/j.chemgeo.2024.122378
7. **Kutyrev, A.**, Kamenetsky, V.S., Chayka, I.F., Nekrylov, N.A., Kryuchkova, L., Shilovskikh, V.V., Kontonikas-Charos, A., Krasheninnikov, S.P., Sapegina, A., Perchuk, A.L., 2024. Metamorphic origin of large nuggets of platinum-group metals: evidence from multiphase inclusions in Os-Ir-Ru alloys from the Adamsfield placer, Tasmania. *Journal of Petrology*. https://doi.org/10.1093/petrology/egae086
8. Zhegunov, P.S., **Kutyrev, A.V.**, Zhitova, E.S., Moskaleva, S.V., Schweigert, P.E., 2024. First data on the mineralogy of the Evevpenta epithermal silver-gold ore occurrence, Kamchatka, Russia. *Journal of Volcanology and Seismology* 18, 149-164. https://doi.org/10.1134/S0742046324700489
9. Stepanov, S., Palamarchuk, R., **Kutyrev, A.**, Lepekhina, E., Sharpenok, L., Shagalov, E., Minervina, E., 2024. Nature of perovskite mineralization of silicate-carbonate veins in the margins of Kusinsko-Kopanskaya layered intrusion (South Urals, Russia). *Minerals* 14, 478. https://doi.org/10.3390/min14050478
10. **Kutyrev, A.**, McDonald, A., Tamura, N., Cabri, L.J., Tolstykh, N., Pekov, I., 2023. Sidorovite, $PtFe_3$, a new mineral from the Snegovaya River placer, Koryak Highlands, Russia. *The Canadian Journal of Mineralogy and Petrology* 61(5), 1021-1030. https://doi.org/10.3749/2200072
11. **Kutyrev, A.**, Kamenetsky, V.S., Kontonikas-Charos, A., Savelyev, D.P., Yakich, T.Y., Belousov, I.A., Sandimirova, E.I., Moskaleva, S.V., 2023. Behavior of platinum-group elements during hydrous metamorphism: constraints from awaruite (Ni_3Fe) mineralization. *Lithosphere*, 2023(1). https://doi.org/10.2113/2023/lithosphere_2023_126
12. Chayka, I.F., Izokh, A.E., Kamenetsky, V.S., Sokol, E.V., Lobastov, B.M., Kontonikas-Charos, A., Zelenski, M.E., **Kutyrev, A.V.**, Sluzhenikin, S.F., Zhitova, L.M., Shvedov, G.I., Shevko, A.Y., Gora, M.P., 2023. Origin of chromitites in the Norilsk-1 intrusion (Siberian LIP) triggered by assimilation of argillaceous rocks by Cr-rich basic magma. *Lithos* 454-455. https://doi.org/10.1016/j.lithos.2023.107254
13. Eslami, A., Grieco, G., Bussolesi, M., Ichiyama, Y., Lenaz, D., Skogby, H., **Kutyrev, A.V.**, Cavallo, A., Khedr, M.Z., 2023. Co-occurrence of compositionally variable chromitites in the Sabzevar ophiolite, NE Iran. *Lithos* 446-447. https://doi.org/10.1016/j.lithos.2023.107133
14. Stepanov, S.Y., Palamarchuk, R.S., **Kutyrev, A.V.**, Shilovskikh, V.V., Petrov, S.V., 2023. Mineralogy and geochemistry of platinum-group elements in the zoned mafic-ultramafic intrusions of the Uralian Platinum belt, Russia. *Journal of Geochemical Exploration* 246. https://doi.org/10.1016/j.gexplo.2023.107155
15. Chayka, I.F., Baykov, N.I., Kamenetsky, V.S., **Kutyrev, A.V.**, Pushkarev, E.V., Abersteiner, A., Shcherbakov, V.D., 2023. Volcano-plutonic complex of the Tumrok Range (Eastern Kamchatka): an example of the Ural-Alaskan type intrusion and related volcanic series. *Minerals* 13, 126. https://doi.org/10.3390/min13010126
16. Yakubovich, O., **Kutyrev, A.**, Sidorov, E., Travin, A., 2022. ^{190}Pt-^{4}He dating of platinum mineralization in Ural-Alaskan-type complexes in the Kamchatka region: evidence for remobilization of platinum-group elements. *Mineralium Deposita*. https://doi.org/10.1007/s00126-022-01103-5
17. Bukhanova, D.S., **Kutyrev, A.V.**, Sidorov, E.G., Chubarov, V.M., 2022. First finding of platinum group minerals at the Malmyzh porphyry gold-copper deposit, Khabarovsk Krai, Russia. *Geology of Ore Deposits* 63, 675-681. https://doi.org/10.1134/s1075701521070035

18. Pekov, I.V., Zubkova, N.V., Agakhanov, A.A., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Sidorov, E.G., **Kutyrev, A.V.**, Pushcharovsky, D.Y., 2022. New arsenate minerals from the Arsenatnaya fumarole, Tolbachik volcano, Kamchatka, Russia. XIX. Axelite, $Na_{14}Cu_7(AsO_4)_8F_2Cl_2$. *Mineralogical Magazine*. https://doi.org/10.1180/mgm.2022.120

19. Pekov, I.V., Zubkova, N.V., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Agakhanov, A.A., Turchkova, A.G., Sidorov, E.G., **Kutyrev, A.V.**, Blatov, V.A., Pushcharovsky, D.Y., 2022. Nishanbaevite, $KAl_2O(AsO_4)(SO_4)$, a new As/S-ordered arsenate-sulfate mineral of fumarolic origin. *Mineralogy and Petrology*. https://doi.org/10.1007/s00710-022-00803-0

20. Cui, M.-M., Su, B.-X., Wang, J., Wu, Y., **Kutyrev, A.**, 2022. Early Paleozoic subduction fingerprints of the Paleo-Asian ocean in easternmost Central Asian Orogenic Belt (CAOB): Identification of the oldest Alaskan-type complex in the CAOB. *GSA Bulletin*. https://doi.org/10.1130/B36307.1

21. Nekrylov, N., Kamenetsky, V.S., Savelyev, D.P., Gorbach, N.V., Kontonikas-Charos, A., Palesskii, S.V., Shcherbakov, V.D., **Kutyrev, A.V.**, Savelyeva, O.L., Korneeva, A.A., Kozmenko, O.A., Zelenski, M.E., 2022. Platinum-group elements in Late Quaternary high-Mg basalts of eastern Kamchatka: Evidence for minor cryptic sulfide fractionation in primitive arc magmas. *Lithos*. https://doi.org/10.1016/j.lithos.2022.106608

22. **Kutyrev, A.V.**, Kamenetsky, V.S., Park, J.-W., Maas, R., Demonterova, E.I., Antsiferova, T.N., Ivanov, A.V., Hwang, J., Abersteiner, A., Ozerov, A.Y., 2021. Primitive high-K intraoceanic arc magmas of Eastern Kamchatka: Implications for Paleo-Pacific tectonics and magmatism in the Cretaceous. *Earth-Science Reviews* 220, 1-22. https://doi.org/10.1016/j.earscirev.2021.103703

23. **Kutyrev, A.**, Zelenski, M., Nekrylov, N., Savelyev, D., Kontonikas-Charos, A., Kamenetsky, V.S., 2021. Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano. *Frontiers in Earth Science* 9. https://doi.org/10.3389/feart.2021.791465

24. **Kutyrev, A.V.**, Sidorov, E.G., Kamenetsky, V.S., Chubarov, V.M., Chayka, I.F., Abersteiner, A., 2021. Platinum mineralization and geochemistry of the Matysken zoned Ural-Alaskan type complex and related placer (Far East Russia). *Ore Geology Reviews* 130, 1-17. https://doi.org/10.1016/j.oregeorev.2020.103947

25. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Agakhanov, A.A., Sandimirova, E.I., Vymazalová, A., Chubarov, V.M., Zolotarev, A.A., 2021. Kufahrite, PtPb, a new mineral from Ledyanoy Creek placer, Galmoenan ultramafic complex, Koryak Highlands, Russia. *Mineralogical Magazine* 85, 254-261. https://doi.org/10.1180/mgm.2021.18

26. Stepanov, S.Y., **Kutyrev, A.V.**, Lepekhina, E.N., Sharpenok, L.N., Antonov, A.V., Kutyreva, M.E., 2021. Age of the dike complex in the dunite "core" of the Kamenushinsky clinopyroxenite-dunite massif, Ural Platinum belt, Middle Urals. *Geochemistry International* 59, 559-576. https://doi.org/10.1134/s0016702921060094

27. Sidorov, E., **Kutyrev, A.**, Chubarov, V., Zhitova, E., 2021. Platinum mineralization of the Epilchik Ural-Alaskan type zoned complex (Far East Russia). *Mineralium Deposita* 56, 143-160. https://doi.org/10.1007/s00126-020-01008-1

28. Ismagilova, R.M., Zhitova, E.S., Krivovichev, S.V., Sergeeva, A.V., Nuzhdaev, A.A., Anikin, L.P., Krzhizhanovskaya, M.G., Nazarova, M.A., Kupchinenko, A.N., Zolotarev, A.A., **Kutyrev, A.V.**, Bukhanova, D.S., Kuznetsov, R.A., Khanin, D.A., 2021. Phase evolution from volborthite, $Cu_3(V_2O_7)(OH)_2{\cdot}2H_2O$, upon heat treatment. *Minerals* 11, 1312. https://doi.org/10.3390/min11121312

29. Saveliev, D.E., Shilovskikh, V.V., Sergeev, S.N., **Kutyrev, A.V.**, 2021. Chromian spinel neomineralisations and the microstructure of plastically deformed ophiolitic peridotites (Kraka massifs, Southern Urals, Russia). *Mineralogy and Petrology* 115, 411-430. https://doi.org/10.1007/s00710-021-00748-w

30. **Kutyrev, A.V.**, Kamenetsky, V.S., Sidorov, E.G., Abersteiner, A., Chubarov, V.M., 2020. Silicate inclusions in isoferroplatinum: Constraints on the origin of platinum mineralization in podiform chromitites. *Ore Geology Reviews* 119, 1-13. https://doi.org/10.1016/j.oregeorev.2020.103367

31. Chayka, I.F., Sobolev, A.V., Izokh, A.E., Batanova, V.G., Krasheninnikov, S.P., Chervyakovskaya, M.V., Kontonikas-Charos, A., **Kutyrev, A.**, Lobastov, B., Chervyakovskiy, V., 2020. Fingerprints of Kamafugite-Like Magmas in Mesozoic Lamproites of the Aldan Shield: Evidence from Olivine and Olivine-Hosted Inclusions. *Minerals* 10, 1-30. https://doi.org/10.3390/min10040337

32. Stepanov, S.Y., Shagalov, E.S., Palamarchuk, R.S., **Kutyrev, A.V.**, Sharpenok, L.N., Nabiullin, F.M., Troshkina, A.N., 2020. Mineralogical and geochemical evidence for granite metasomatism in dikes in the north of the Berezovskoe Ore field (Middle Urals). *Russian Geology and Geophysics* 61, 809-827. https://doi.org/10.15372/RGG2019176

33. **Kutyrev, A.V.**, Zhirnova, T.S., 2019. Concentrically-zoned massifs of the Tamanvayam group (Koryak-Kamchatka Platiniferous Belt): structure, age, petrological and geochemical aspects. *Russian Journal of Pacific Geology* 13, 350-363. https://doi.org/10.1134/S1819714019040031

34. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Chubarov, V.M., Khanin, D.A., 2019. Origin of platinum-group mineral assemblages from placers in rivers draining from the Ural-Alaskan type Itchayvayamsky ultramafics, Far East Russia. *Canadian Mineralogist* 57, 91-104. https://doi.org/10.3749/canmin.1800040

35. **Kutyrev, A.V.**, Sidorov, E.G., Antonov, A.V., Chubarov, V.M., 2018. Platinum-group mineral assemblage of the Prizhimny Creek (Koryak Highland). *Russian Geology and Geophysics* 59, 935-944. https://doi.org/10.1016/j.rgg.2018.07.014

Under review

1. **Kutyrev, A.**, Maier, W., Karinen, T., Barnes, S.-J., Hayes, B., Savard, D., Virtanen, V., Booth, C., Muir, D. Petrogenesis of the Akanvaara layered intrusion, Finnish Lapland. *Journal of Petrology*, under review.

2. **Kutyrev, A.**, Kamenetsky, V.S., Chayka, I.F., Yudovskaya, M.A., Eslami, A., Kislov, E.V., Pushkarev, E.V., Zhitova, L.M. What do Cr-spinel-hosted multiphase inclusions in chromitites record? *Journal of Petrology*, under review.

3. Booth, C.A., Hu, H., Jackson, M.D., Virtanen, V.J., **Kutyrev, A.V.**, Maier, W.D. Reactive flow drives the compositional evolution of igneous rocks in the Earth's crust. Under review.

4. Chayka, I., Hu, W.-J., **Kutyrev, A.**, Petrenko, D., Zhitova, L., Shevko, A., Gora, M., Kalugin, V., Shvedov, G., Izokh, A., Yudovskaya, M. Fluid-mediated PGE enrichment of magmatic low-sulfide ores as modelled for the Norilsk deposits. *Geology*, under review.

5. Baykov, N.I., Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Nekrylov, N.A., Shcherbakov, V.D., Zhukova, I.A., Plechov, P.Yu., Abramova, V.D., Stepanov, A.S. Melt-spinel interface effects control Pt fractionation in basaltic melt. *Geology*, under review.

Datasets and code

1. **Kutyrev, A.V.**, Maier, W.D., Karinen, T., Barnes, S.J., Savard, D., 2025. Whole-rock geochemical dataset of the Akanvaara layered intrusion, Lapland, Finland. Dataset deposited with the Interdisciplinary Earth Data Alliance (IEDA). https://doi.org/10.60520/IEDA/113569

2. Filimonova, O., **Kutyrev, A.**, Wermeille, D., 2025. Iridium L_3-edge XAS dataset of natural Ir-O phase, comprising IrO_2 [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.15650871

3. Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Baykov, N.I., Barnes, S., Savard, D., Alexei, I.V., 2025. Major, trace, and platinum-group elements in the Late Cretaceous arc volcano-plutonic complex of the Tumrok Range, eastern Kamchatka, Version 1.0. Interdisciplinary Earth Data Alliance (IEDA). https://doi.org/10.60520/IEDA/113578

4. **Kutyrev, A.**, Bindeman, I., O'Neil, J., Rizo, H., 2024. Code for 1- and 2-stage oxygen fractionation and metamorphic dehydration modeling from "Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada". https://doi.org/10.5281/zenodo.12795578

Conference presentations

1. IMSG 2026, Barberton, South Africa. Oral presentation: "Mottled anorthosites and discordant ultramafic pipes in the MG4-UG1 interval of the Upper Critical Zone, Bushveld Complex".

2. Geocongress 2025, Bloemfontein, South Africa. Oral presentation: "Petrogenesis of the Akanvaara layered intrusion".

3. MDSG 2024, Edinburgh. Poster: "Assessing mass-balance of layered intrusions: the Akanvaara intrusion case study in Lapland, Finland".

4. Goldschmidt 2023, Lyon, France. Remote oral talk: “Early Earth's hydrosphere: insights from the hydrothermally altered submarine lavas of the Eoarchean Saglek-Hebron Complex, Canada”.
5. 14th International Platinum Symposium 2023, Cardiff, UK. In-person oral presentation: “Origin of native osmium in metamorphosed peridotites: insights from multiphase inclusions (Adamsfield placer, Tasmania)”.
6. Virtual Early Career Platinum Symposium 2022. Oral presentation: “Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano, Kamchatka”.
7. EGU General Assembly 2021. Oral presentation: “Awaruite (Ni_3Fe) as a platinum-group elements concentrator: Preliminary data”.
8. SGA 2019 Conference, Glasgow, United Kingdom. In-person oral presentation: “Platinum mineralization in Ural-Alaskan complexes of Koryak Highlands (Far East Russia)”.